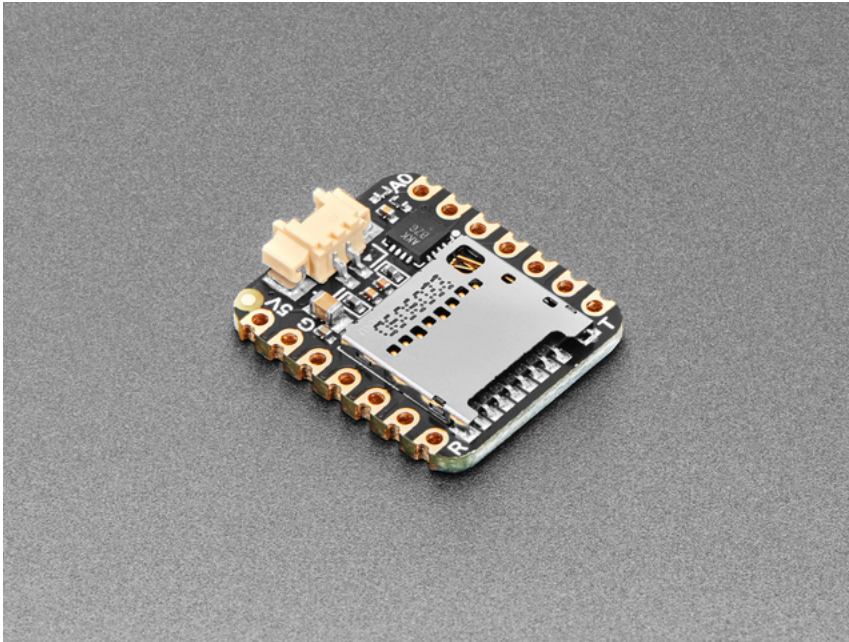


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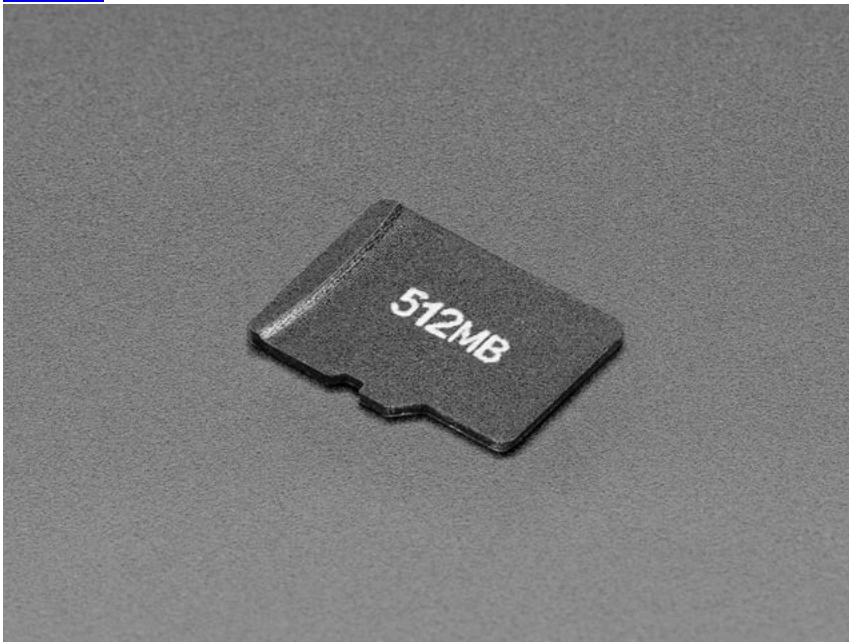


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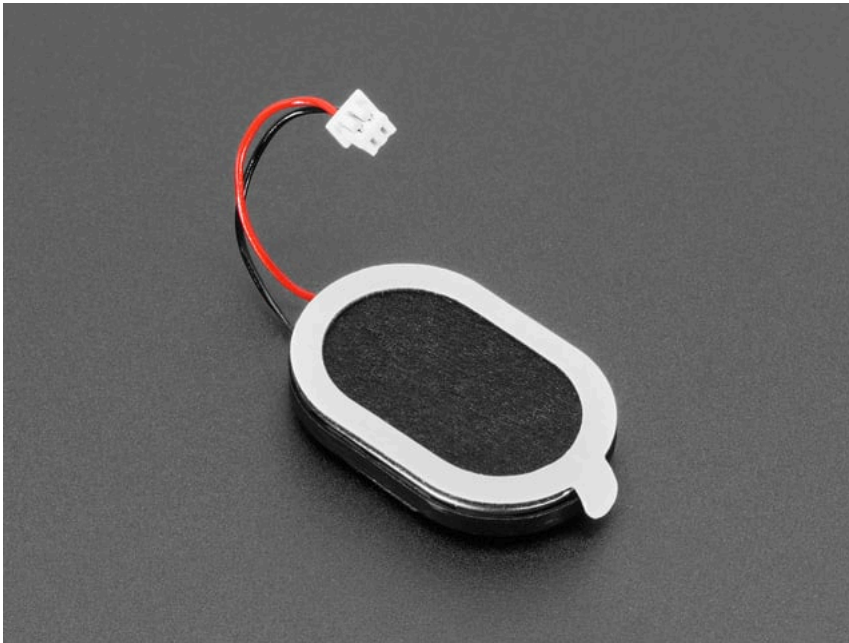


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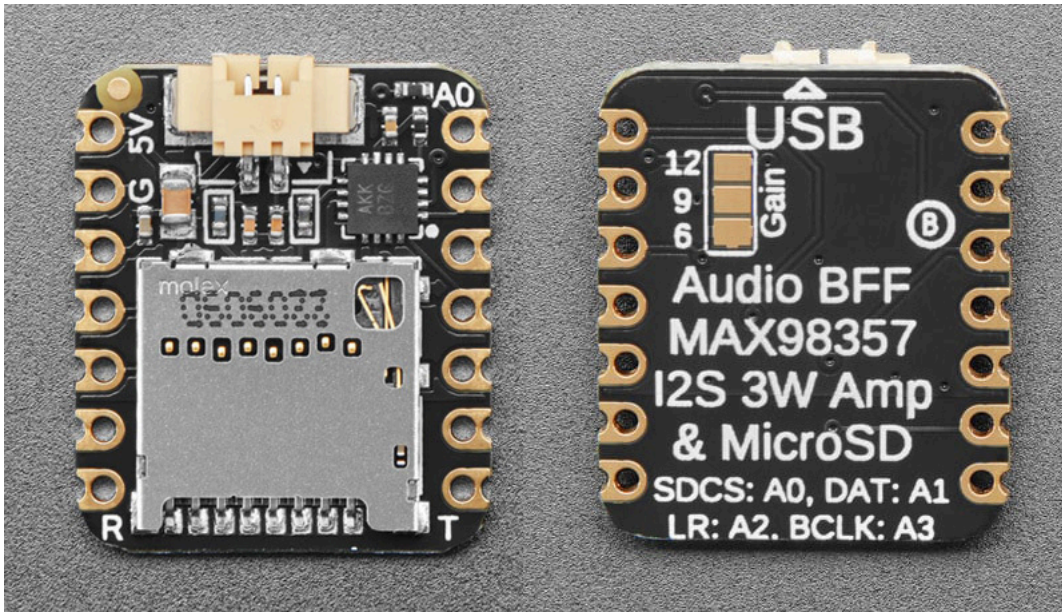
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Beginner

Product guide

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Pinouts



Text emphasized with a blue exclamation: This breakout is meant to be used with a QT Py or Xiao form factor board. Using with other form factor boards may be difficult.

I2S Amplifier

- The grey square located towards the top right of the BFF is the **MAX98357 I2S amplifier chip**. It supports digital audio.

Default I2S Pins

These pins are compatible with ESP32-series, nRF52840 and RP2040 microcontroller boards.

- **A1** - This is used for audio data, or **DAT**.
- **A2** - This is used for wordselect clock, or **LR**.
- **A3** - This is used for bitclock, or **BCLK**.



Text emphasized with a yellow exclamation: These pins will not work with the QT Py SAMD21 microcontroller board.

Default SD Card Pins

The SD Card is connected to the default SPI pins on the QT Py. **SDCS** is connected to pin **A0**.

- **MO** - This is the SPI MOSI (**M**icrocontroller **O**ut / **S**erial **I**n) pin. It is used to send data from the microcontroller to the SD card. It is connected to the default **MOSI** pin on the QT Py.
- **MI** - This is the SPI MISO (**M**icrocontroller **I**n / **S**erial **O**ut) pin. It's used for sending data from the SD card to the microcontroller. It is connected to the default **MISO** pin on the QT Py.
- **SCK** - This is the SPI clock input pin. It is connected to the default **SCK** pin on the QT Py.
- **A0** - This is the SD chip select pin (**SDCS**).

I2S Gain Jumper

On the back of the board, towards the top, are the I2S gain jumper pads. They are outlined in white on the board silk and are labeled **Gain**. The default gain for the amplifier is 9 dB.

- To set the gain to 6 dB, solder the center jumper, labeled **9**, to the jumper above labeled **6**. This connects the **GAIN** pin on the I2S amplifier to **5V**.
- To set the gain to 12 dB, solder the center jumper, labeled **9**, to the right jumper below labeled **12**. This connects the **GAIN** pin on the I2S amplifier to **GND**.

Speaker Connector

- This **two-pin PicoBlade-compatible speaker connector**, located at the center of the front of the BFF, is meant for plugging in a **4-8Ω, 3W-or-less speaker**.
- To connect a speaker with bare wires, you'll want to pick up [this cable](#).
- If you need a speaker, [this speaker](#) is an excellent place to start.

micro SD Card Slot

- On the front of the board, towards the bottom, is the micro SD card slot. You can use any micro SD card that supports SPI mode with one CS pin.

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